

Electroluminescence Yield for $\lambda > 165$ nm in Neon-Xenon Mixtures: Experimental Results

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Abstract

The electroluminescence yield for $\lambda > 165$ nm in neon-xenon mixtures at a total pressure of about 800 torr, is studied as a function of the reduced electric field in a uniform-field gas proportional scintillation counter. Scintillation and ionization thresholds were experimentally observed to decrease monotonically as a function of xenon concentration, from approximately 1 and 6 Vcm⁻¹ torr⁻¹ for 100% xenon, to 0.5 and 3.8 Vcm⁻¹ torr⁻¹ for 20% xenon, to 0.4 and 2.8 Vcm⁻¹ torr⁻¹ for 10% xenon and to 0.3 and 2.2 Vcm⁻¹ torr⁻¹ for 5% xenon. Detector energy resolutions for the aluminum K α line at 1.487 keV were determined to be 15% for 20% xenon, 19% for 10% xenon and 22% for 5% xenon mixtures.

shift the electroluminescence spectra to that characteristic of pure xenon [3,4], we decided to study the potential of neon-xenon mixtures as filling gases for X-ray detection. Helium based mixtures were eliminated since this gas would permeate the detector entrance window as well as the quartz window of the PMT.

In this work we present a study of the reduced electroluminescence yield of neon-xenon mixtures (100, 20, 10, and 5% Xe) at total pressures of about 800 torr, as a function of the reduced electric field. Scintillation thresholds are determined and ionization thresholds are estimated. This study is essential for the calculations required for the design of large area GPSCs using the curved grid technique [5,6].

I. INTRODUCTION

Applications of the gas proportional scintillation counter (GPSC) include X-ray astronomy and X-ray fluorescence analysis instrumentation. The advantage of the GPSC over other radiation detectors is derived from the amplification of the primary electron signal by the production of vacuum ultraviolet (VUV) electroluminescence whose spectrum is peaked at 170 nm in xenon. Xenon gas is preferred since it has a large X-ray photoionization cross section and because its electroluminescence spectrum can be detected with a photomultiplier tube (PMT) having a high-purity quartz window (about 50% transmission at 165 nm).

At X-ray energies below 2 or 3 keV, however, the measured X-ray pulse-height distribution in a xenon-filled GPSC departs from a gaussian shape with the observation of a low-energy tail. This distortion was explained in detail by Monte Carlo simulation[1,2] as being due to a partial loss of primary electrons to the detector entrance window. This effect can be reduced to some degree by increasing the electric field near the entrance window, but this method is limited due to the onset of unwanted scintillation and ionization in the absorption region of the detector [1,2].

An alternative solution might be to use noble gases with a lower X-ray cross section so that X-ray absorptions occur, on the average, further away from the entrance window. Obvious choices of helium and neon were eliminated because of their low electroluminescence yields at wavelengths detectable with quartz window PMTs. However, as small amounts of xenon introduced at atmospheric pressure in the lighter noble gases,

II. EXPERIMENTAL METHOD

The uniform field GPSC described in [7] was used for the measurements presented here. The detector has a 25- μ m thick Kapton film entrance window and is instrumented with a high-purity quartz window PMT (EMI D676QB). The GPSC drift and scintillation regions are 4- and 1-cm deep, respectively. The reduced electric field in the drift region was maintained constant at 0.1 V cm⁻¹ torr⁻¹. The PMT was polarized at 800 V and its anode signals were processed by an ORTEC 121 preamplifier, followed by a HP5582A amplifier. Pulse-height distributions were obtained with a Wilkinson type multichannel analyzer.

Pulse-height distributions of a collimated 5.9 keV X-ray beam from an ⁵⁵Fe source (with the 6.4 keV K β line removed by a chromium filter) were measured as a function of gas mixtures and electric field. To reduce errors due to ballistic effects, 5 μ s integration and differentiation time constants were used. The linearity of the amplification and multichannel analyzer stages was checked with a precision BNC PB-3 pulse generator.

Each 5.9 keV X-ray pulse-height distribution was fitted to a gaussian curve superimposed on a linear background using the grid-search least-squares (GRIDLS) method[8]. The centroid and full width at half maximum (FWHM) were obtained. The channel number of the centroid is proportional to that part of the light yield measured with the D676QB PMT, namely for $\lambda > 165$ nm, which, in turn, is proportional to the total light yield, only if the electroluminescence spectral distribution remains unchanged as the gas mixtures and electric field intensities vary.

III. RESULTS

For each neon-xenon mixture we obtained the centroid channel number of the pulse-height distribution and the detector energy resolution as a function of the reduced electric field (E/p) in the scintillation region. The centroid channel number divided by the total gas pressure is proportional to the reduced electroluminescence yield (i.e., the number of photons, n , produced by one electron drifting the unit of distance, x , divided by the gas pressure, $\frac{1}{p} \cdot \frac{dn}{dx}$) for

$\lambda > 165$ nm. In Figure 1 we present the non-normalized reduced electroluminescence yield for $\lambda > 165$ nm as a function of the reduced electric field in the scintillation region, for the different relative neon-xenon concentrations (100, 20, 10 and 5% Xe). However, we must point out that since the W -value, and so the number of primary electrons, varies with the neon-xenon relative concentration, the different curves represented in this figure do not have the same normalization. Actually, what is represented in the vertical axes is the electroluminescence intensity due to a single 5.9 keV X-ray photon, divided by the total gas pressure.

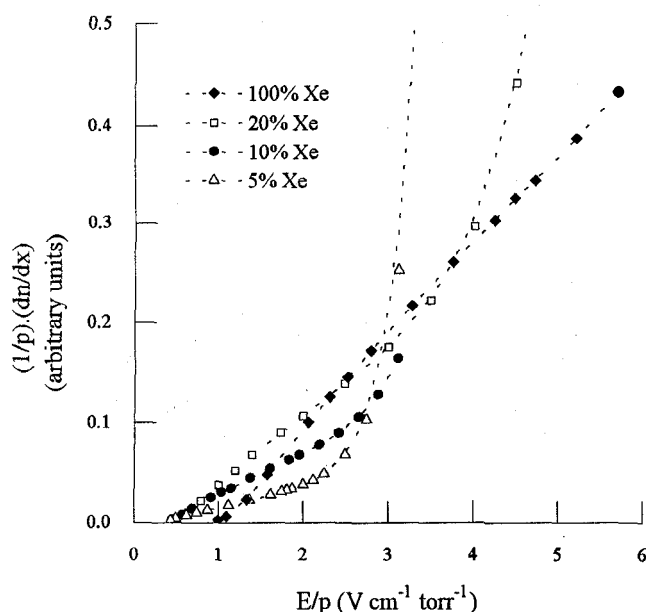


Figure 1 - Non-normalized reduced electroluminescence yield for $\lambda > 165$ nm (centroid channel number divided by total gas pressure) as a function of the reduced electric field (electric field divided by total gas pressure) for different neon-xenon concentrations.

As shown the thresholds for scintillation decrease with decreasing xenon concentration and are well below the $1 \text{ V cm}^{-1} \text{ torr}^{-1}$ threshold already obtained for pure xenon[9]. These thresholds are 0.5 for 20% xenon, 0.4 for 10% xenon and 0.3 for 5% xenon. This effect can be explained in terms of a balance of the energy gained by electrons from the electric field between collisions and the energy lost during elastic collisions. The energy gain depends on the elastic

cross-sections for the gas components and on their concentrations. The energy lost by electrons (mass m) in elastic collisions with neutral atoms (mass M), is about $2 m/M$. Thus, as the total collision cross-section for electrons in neon is rather smaller than in xenon [10,11], electrons drifting in neon-xenon mixtures undergo a smaller number of collisions than in pure xenon and, therefore, the energy gain from the field is accordingly larger. On the other hand, the energy lost by electrons in elastic collisions with neon atoms is 6.5 times larger than with xenon atoms.

From the experimental results shown in Figure 1, we can conclude that the energy gain due to the lower neon cross-sections is the stronger of the two effects since, as xenon concentration is lowered, the threshold for scintillation decreases. Alternatively, this is equivalent to saying that electrons behave as they would in xenon gas at lower pressure under the same electric field. But, as what is plotted in Fig. 1 is the electroluminescence yield as a function of the electric field divided by the total gas pressure (and not divided by the partial xenon pressure), the scintillation threshold shifts to the left.

Figure 1 also shows that above the scintillation thresholds, the reduced electroluminescence yield decreases with decreasing xenon concentration. This can be explained in terms of a shift towards shorter wavelengths of the xenon electroluminescence at lower pressure, which is then due to the xenon first continuum emission[12], peaked at about 150 nm. This emission is not transmitted through the PMT window. Another reason for this decrease is the likely reduction in the number of primary electrons for the mixtures with a large amount of neon, since the W -value for pure neon is almost a factor of 2 larger than that for pure xenon. These effects, leading to a decrease in the number of primary electrons and in the amount of light detected by the PMT, might be responsible for the deterioration of the energy resolution which is for 5.9 keV X-rays, 7.7% for 100% xenon, 9.0% for 20% xenon, 10.0% for 10% xenon and 12.5% for 5% xenon.

Above the scintillation thresholds, the electroluminescence yields demonstrate a linear behavior until a faster rising region is encountered. This region can be explained by the onset of ionization. From Figure 1, we can then estimate that the ionization thresholds are about $3.8 \text{ V cm}^{-1} \text{ torr}^{-1}$ for 20% xenon, $2.8 \text{ V cm}^{-1} \text{ torr}^{-1}$ for 10% xenon and $2.2 \text{ V cm}^{-1} \text{ torr}^{-1}$ for 5% xenon. The same explanation that was presented above for the balance of the energy gained by electrons from the electric field between collisions and the energy lost during elastic collisions can be invoked to explain the decrease in the threshold for ionization with decreasing xenon concentration.

Figure 2 shows the pulse-height distributions obtained for the aluminum K_{α} fluorescent X-ray from the excitation of an aluminum foil with α particles, for the three different neon-xenon mixtures.

The main peak (channel number around 700) corresponds to the Al K_{α} line (1.487 keV) - the lowest energy

transmitted through the Kapton window. The higher energy peak (channel number around 1400) corresponds to impurities, either from the radioactive source or from the fluorescence of argon in the atmospheric air.

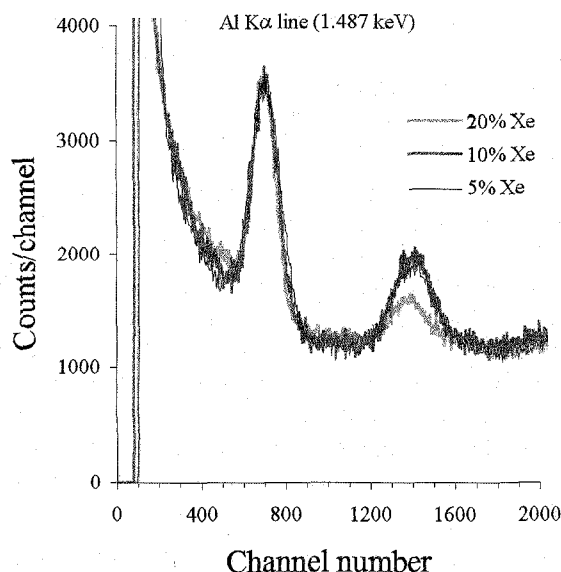


Figure 2 - Pulse-height distributions for the Al K α line (1.457 keV) for different neon-xenon concentrations.

The measured energy resolutions were 15% for 20% xenon, 19% for 10% xenon and 22% for 5% xenon. Although there was no improvement in the energy resolution due to the reasons explained above, we expected that the tail would have a more significant decrease with decreasing xenon concentration than the one presented in Figure 2. A more detailed study concerning the behavior of these neon-xenon mixtures for lower energy X-rays, as has already been performed for the case of pure xenon[2], may be of interest.

IV. CONCLUSIONS

The electroluminescence threshold for neon-xenon mixtures was found to decrease with decreasing xenon concentration. This was explained in terms of the neon/xenon mass ratio and the electron elastic collision cross-sections. Above the scintillation threshold, linear behavior is exhibited until the ionization threshold is reached. However, the relative intensity of the light detected by the PMT decreases with xenon concentration, an effect attributed to both the shift towards shorter wavelengths of the electroluminescence and to the expected increase in the W-value for mixtures with large amounts of neon. Although recent results[13] have shown that the shifting to shorter wavelengths is small, work with MgF₂ window PMT is already in progress. The measured energy resolution for 5.9 keV X-rays also deteriorates with xenon concentration (from 7.6% in 100% xenon to 12.5% in 5% xenon); the expected decrease in the tail for the lowest X-ray energy studied was not clearly observed.

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